

Incomplete Markets with Hand-to-Mouth Agents I will now relax the assumption that agents are not allowed to borrow and save by allowing them access to a risk-free bond. To ensure that the problem remains tractable, I will restrict the tax function (HDtax) by assuming that the history dependence parameters take the form $\theta_j = (1 - \beta\mu)\mu^j$, $j \geq 1$.

That is, the tax function takes the form in Proposition (mainprop), with the added assumption that both roots are the same, $\mu_1 = \mu_2 = \mu$. The promise-keeping constraint (omeganorm) restricts the initial value θ_0 to $1 - \beta\mu$, already incorporated in the tax function above. The history-independent tax system sets $\mu = 0$. The above representation again allows us to write the tax function recursively by align*

$$T(y_t) = y_t - \lambda(y_t m_t)^{(1-\tau)\theta_0}, \quad m_{t+1} = (y_t m_t)^\mu.$$